

Ratio Toolkit

Two methods solve every ratio question: the Multiplier for recipes and scaling, and the Value of 1 Part for sharing.



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THE TOPIC ON ONE MAP

Ratio Toolkit

01 Recipes and scaling: the Multiplier

10 pancakes → 30

enough flour?

greatest number of biscuits

02 Simplify and $1 : n$

2 cm : 5 m

720 : 540

map scale $1 : n$

03 Sharing: the Value of 1 Part

£630 in $5 : 13$

12 more goals

36 pigs

04 Linking two ratios

$7 : 2$ and $5 : 9$

flour : butter : sugar

05 Ratio problems

coins worth 85p

counters added

best buy of paint

VALUE OF 1 PART Three ways to find one part — the question tells you which:

ZONE A

Given the TOTAL

$$1 \text{ part} = \frac{\text{total}}{\text{sum of parts}}$$

ZONE B

Given a DIFFERENCE

$$1 \text{ part} = \frac{\text{difference}}{\text{diff. in parts}}$$

ZONE C

Given ONE SHARE

$$1 \text{ part} = \frac{\text{that share}}{\text{its parts}}$$

01 Recipes and Scaling: the Multiplier

Find the Multiplier = $\frac{\text{new}}{\text{old}}$, then multiply every ingredient. For “greatest number”, find each ingredient’s multiplier — the **smallest** one wins.

WHEN YOU SEE → YOU DO

“ingredients for 16 ... he makes 24”

→ multiplier $\frac{24}{16} = 1.5$

“does she have enough flour?”

→ amount needed vs amount she has

“greatest number she can make”

→ each ingredient: have ÷ need — take the smallest

“160 cm weighs 17.8 g ... 210 cm”

→ $17.8 \times \frac{210}{160}$

Multiplier = new ÷ old

× every amount

Units and rounding

WORKED MODEL • GREATEST NUMBER

8 shortbread use 120 g butter, 60 g sugar, 180 g flour. Tariq has 330 g, 200 g, 450 g.

1 $330 \div 120 = 2.75$, $200 \div 60 = 3.3$, $450 \div 180 = 2.5$

each multiplier

2 smallest = 2.5 (flour runs out first)

the limit

ANSWER $2.5 \times 8 = 20$ biscuits

IDEA 01

Practise: Recipes and Scaling: the Multiplier

My questions first, then selected practice.

Q1 DR ESLAM Multiplier

CH 1 • TOPIC 12 2 marks

A recipe for 6 people uses 450 g of rice. How much rice for 10 people?

SELECTED PRACTICE

Q2**PRACTICE** Practice • Easy

CH 1 • TOPIC 12 4 marks

- 1 (a) Here are the ingredients needed to make 10 pancakes.

PancakesIngredients to make **10** pancakes300 *ml* of milk

120 g of flour

2 eggs

Matthew makes 30 pancakes.

Work out how much flour he uses.

(2 marks)

- (b) Tara makes some pancakes.

She uses 750 *ml* of milk.

Work out how many pancakes she makes.

(2 marks)

Q3**PRACTICE**

Practice • Easy

CH 1 • TOPIC 12

3 marks

- 3 This is a list of ingredients for making chicken soup for 4 people.

Ingredients for **4** people

60 g butter
300 g chicken
150 ml cream
1 onion
640 ml chicken stock

Bill is going to make chicken soup for 6 people.

Work out the amount of each ingredient he needs.

(3 marks)

Q4**PRACTICE**

Practice • Easy

CH 1 • TOPIC 12**3 marks**

12 18 rice cakes weigh a total of 130 g. There are 329 calories in 100 g of rice cakes.

How many calories are there in one rice cake?

..... calories

(3 marks)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 12

4 marks

2 Here is a list of ingredients for making 18 mince pies.

Ingredients for mince pies

225 g of butter
350 g of flour
100 g of sugar
280 g of mincemeat
1 egg

Elaine wants to make 45 mince pies.

Elaine has

1 kg of butter
1 kg of flour
500 g of sugar
600 g of mincemeat
6 eggs

Does Elaine have enough of each ingredient to make 45 mince pies?
You must show clearly how you got your answer.

(4 marks)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 12 3 marks

5 There are 1200 students at a school.

Kate is helping to organise a party.
She is going to order pizza.

Kate takes a sample of 60 of the students at the school.
She asks each student to tell her **one** type of pizza they want.

The table shows information about her results.

Pizza	Number of students
ham	20
salami	15
vegetarian	8
margarita	17

Work out how much ham pizza Kate should order.

Write down any assumption you make **and** explain how this could affect your answer.

(3 marks)

Q7**PRACTICE**

Practice • Hard

CH 1 • TOPIC 12

5 marks

- 1 Colin works on 5 days each week.
Each day he drives from his home to work and from work to his home.

Colin pays £3.50 each day to use the car park at work.

The distance from Colin's home to work is 18 miles.
Colin's car uses one gallon of petrol every 45.2 miles.

1 litre of petrol costs 136.9p

1 gallon = 4.546 litres

Work out the total cost for Colin to use his car for work each week.
You must show all your working.

(5 marks)

02 Simplify and $1 : n$

Same units first, then divide both sides by the same number. For $1 : n$, divide both sides by the first number.

WHEN YOU SEE → YOU DO

“2 cm to 5 m in the form $1 : n$ ”

→ same units: $2 : 500 \rightarrow 1 : 250$

“in its simplest form”

→ divide by the HCF: $720 : 540 = 4 : 3$

“A is half of B”

→ $A : B = 1 : 2$

“scale $1 : 200$... length of the model”

→ real $\div 200$

WORKED MODEL • $1 : n$

Write 50 grams : 1 kilogram in the form $1 : n$.

1 50 : 1000

same units

ANSWER 1 : 20

× **TRAP** $2 \text{ cm} : 5 \text{ m} = 2 : 5$ is wrong — change metres to centimetres first.

IDEA 02

Practise: Simplify and $1 : n$

My questions first, then selected practice.

Q8 DR ESLAM $1 : n$

CH 1 • TOPIC 12 2 marks

Write $40 \text{ cm} : 2 \text{ m}$ in the form $1 : n$.**SELECTED PRACTICE****Q9** PRACTICE Practice • Easy

CH 1 • TOPIC 12 1 mark

9 A is half of B .Work out the ratio $A : B$

Circle your answer.

1 : 2	2 : 1	1 : 3	3 : 1
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(1 mark)

Q10**PRACTICE**

Practice • Easy

CH 1 • TOPIC 12 2 marks

- 11 The ratio 50 grams to 1 kilogram can be written in the form $1 : n$.

Find the value of n .

$n = \dots\dots\dots$

(2 marks)

Q11**PRACTICE**

Practice • Medium

CH 1 • TOPIC 12 4 marks

11 (a) Emma has a digital photo.

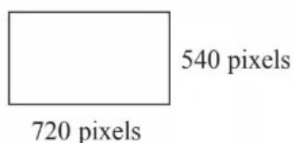


Diagram **NOT**
accurately drawn

The photo has a width of 720 pixels.

The photo has a height of 540 pixels.

Write down the ratio of the width of the photo to the height of the photo.

Give your ratio in its simplest form.

(2 marks)

(b) Emma wants the ratio of the width of the photo to the height of the photo to be 3 : 2

She reduces the number of pixels in the height of the photo.

The width of the photo is still 720 pixels.

The ratio of the width of the photo to the new height of the photo is 3 : 2

Work out the new height of the photo.

(2 marks)

Q12 **PRACTICE** Practice · Medium

CH 1 • TOPIC 12 3 marks

15 A plane has a length of 73 metres.

A scale model is made of the plane.
The scale of the model is 1 : 200

Work out the length of the scale model.
Give your answer in centimetres.

(3 marks)

Q13 **PRACTICE** Practice · Hard

CH 1 • TOPIC 12 3 marks

11 Work out

cube root of 512 : reciprocal of 0.4

Give your answer in the form $n : 1$

(3 marks)

Q14 **PRACTICE** Practice • Very HardCH 1 • TOPIC 12 **3 marks**

- 3 The surface gravity of a planet can be worked out using the formula

$$g = \frac{6.67 \times 10^{-11} m}{r^2}$$

where

m kilograms is the mass of the planet

r metres is the radius of the planet

For the Earth and Jupiter here are the values of m and r .

Earth	Jupiter
$m = 5.98 \times 10^{24}$	$m = 1.90 \times 10^{27}$
$r = 6.378 \times 10^6$	$r = 7.149 \times 10^7$

Work out the ratio of the surface gravity of Earth to the surface gravity of Jupiter.

Write your answer in the form $1 : n$

(3 marks)

03 Sharing: the Value of 1 Part

Step 1: find the value of 1 part (Zone A, B or C). Step 2: multiply by the number of parts you need.

WHEN YOU SEE → YOU DO

“share £630 in the ratio 5 : 13”

→ Zone A: $630 \div 18$

“Simon scored 12 more goals than Arjun”

→ Zone B: $12 \div (5 - 2)$

“95 girls, boys : girls = 4 : 5”

→ Zone C: $95 \div 5$

“the total that Asha and Julie receive is \$36”

→ Zone A on their parts only

Which zone?

Value of 1 part

× parts you need

WORKED MODEL • MAY 2023 P1H

Goals Arjun : Simon : Kath = 2 : 5 : 8. Simon scored 12 more than Arjun. How many did Kath score?

1 $12 \div (5 - 2) = 4$

Zone B: 1 part

ANSWER $8 \times 4 = 32$ goals

IDEA 03

Practise: Sharing: the Value of 1 Part

My questions first, then selected practice, then real past papers.

Q15 DR ESLAM Zone A

CH 1 • TOPIC 12 2 marks

Share £360 in the ratio 2 : 3 : 4.

Q16 DR ESLAM Zone B

CH 1 • TOPIC 12 2 marks

Boys : girls = 5 : 3. There are 18 more boys than girls. How many students?

SELECTED PRACTICE

Q17**PRACTICE**

Practice • Easy

CH 1 • TOPIC 12

5 marks

6 (a) Jon shares £700 equally between his two children, Ellie and Maddie.

Ellie gives £125 of her share of the money to Maddie.

Write down the ratio of the amount of money Ellie now has to the amount of money Maddie now has.

(2 marks)

(b) Jenny shares £630 between her two children, Daniel and Rose, in the ratio 5 : 13

Work out how much money Jenny gives to each child.

(3 marks)

Q18**PRACTICE**

Practice • Medium

CH 1 • TOPIC 12 6 marks

9 (a) Babajan makes breakfast cereal.

She mixes nuts, raisins and oats in the ratio 3 : 2 : 5 by weight.

On Monday, Babajan uses 60 grams of nuts.

Work out the weight of raisins and the weight of oats she uses to make the breakfast cereal.

(3 marks)

(b) On Tuesday, Babajan makes 300 grams of the breakfast cereal.

500 grams of nuts cost £8

Work out the cost of the nuts used to make 300 grams of the breakfast cereal.

(3 marks)

Q19**PRACTICE**

Practice • Hard

CH 1 • TOPIC 12**3 marks**

- 4 Frank, Mary and Seth shared some sweets in the ratio 4 : 5 : 7 Seth got 18 more sweets than Frank.

Work out the total number of sweets they shared.

(3 marks)**Q20****PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 12**3 marks**

- 4 Danil, Gabriel and Hadley share some money in the ratios 3 : 5 : 9
The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros.

Work out the amount of money that Danil receives.

..... euros

(3 marks)

PAST PAPERS

Q21 PAST PAPER Edexcel 4MA1 • May 2023 • Paper 1H • Q1

CH 1 • TOPIC 12 3 marks

You must write down all the stages in your working.

- 1 Last season, the number of goals scored by Arjun, by Simon and by Kath for their football team were in the ratios $2:5:8$

Simon scored 12 more goals than Arjun.

Work out the number of goals scored by Kath.

(Total for Question 1 is 3 marks)

04 Linking Two Ratios

Two ratios share one item. Make that item the **same number** in both (multiply each ratio), then join them into one three-part ratio.

WHEN YOU SEE → YOU DO

“chickens : ducks = 7 : 2, ducks : pigs = 5 : 9”

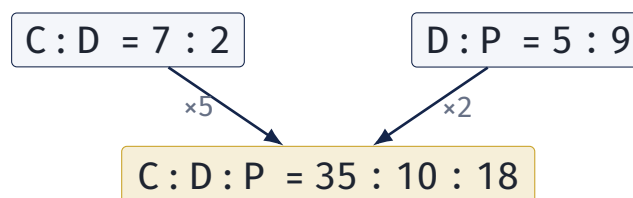
→ ducks → LCM 10: 35 : 10 : 18

“flour : butter = 6 : 5, butter : sugar = 3 : 2”

→ butter → 15: 18 : 15 : 10

“there are 36 pigs”

→ Zone C on the joined ratio



WORKED MODEL • MAY 2021 P1H

Chickens : ducks = 7 : 2, ducks : pigs = 5 : 9, 36 pigs. How many chickens?

1 $35 : 10 : 18$

ducks made equal (10)

2 $36 \div 18 = 2$

Zone C: 1 part

ANSWER $35 \times 2 = 70$ chickens

IDEA 04

Practise: Linking Two Ratios

My questions first, then real past papers.

Q22 DR ESLAM Link

CH 1 • TOPIC 12 3 marks

$a : b = 3 : 4$ and $b : c = 6 : 5$. Write $a : b : c$ in its simplest form.

PAST PAPERS**Q23** PAST PAPER Edexcel 4MA1 • May 2021 • Paper 1H • Q4

CH 1 • TOPIC 12 3 marks

- 4 On a farm there are chickens, ducks and pigs.

The ratio of the number of chickens to the number of ducks is 7:2

The ratio of the number of ducks to the number of pigs is 5:9

There are 36 pigs on the farm.

Work out the number of chickens on the farm.

.....

(Total for Question 4 is 3 marks)

Q24 **PAST PAPER** Edexcel 4MA1 • November 2023 • Paper 2H • Q5**CH 1 • TOPIC 12** **3 marks**

5 Avril bakes a cake.

She uses flour, butter and sugar such that

$$\text{weight of flour : weight of butter} = 6 : 5$$

$$\text{weight of butter : weight of sugar} = 3 : 2$$

Avril uses 120 grams of sugar.

Work out the weight of flour Avril uses.

..... grams

(Total for Question 5 is 3 marks)

05 Ratio Problems

The ratio counts things, but the question may be about value, weight or cost. Use the parts, then turn them into what the question asks.

WHEN YOU SEE → YOU DO

“2p and 5p coins in ratio 1 : 3, total 85p”

→ value of one group: $1 \times 2 + 3 \times 5 = 17p$

“she puts more red counters in ... now 1 : 2”

→ yellow stays the same — find new red, subtract

“ratio now 10 : 11 — more than 90 people?”

→ list the possible totals

containers of paint, “make 40% profit”

→ cost per litre of the mix, then $\times 1.4$

WORKED MODEL • JUNE 2023 P2H

Nancy has 2p and 5p coins in ratio 1 : 3, total 85p. How many more 5p coins than 2p?

1 one group: $1 \times 2p + 3 \times 5p = 17p$

value of one group

2 $85 \div 17 = 5$ groups: 5 two-pence, 15 five-pence

number of groups

ANSWER $15 - 5 = 10$ more

IDEA 05

Practise: Ratio Problems

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE**Q25****PRACTICE**

Practice • Medium

CH 1 • TOPIC 12 4 marks

8 5 schools sent some students to a conference.

One of the schools sent both boys and girls.

This school sent 16 boys.

The ratio of the number of boys it sent to the number of girls it sent was 1:2

The other 4 schools sent only girls.

Each of the 5 schools sent the same number of students.

Work out the total number of students sent to the conference by these 5 schools.

(4 marks)

Q26**PRACTICE**

Practice • Hard

CH 1 • TOPIC 12 3 marks

- 12 Purple paint is made by mixing red paint and blue paint in the ratio 5 : 2
Yan has 30 litres of red paint and 9 litres of blue paint.

What is the **maximum** amount of purple paint he can make?

.....litres

(3 marks)

Q27**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 12 3 marks

- 5 The ratio $(y + x) : (y - x)$ is equivalent to $k : 1$

Show that $y = \frac{x(k+1)}{k-1}$

(3 marks)

Q28**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 12 4 marks

Volume of a sphere = $\frac{4}{3} \pi r^3$ where r is the radius

Volume of a cone = $\frac{1}{3} \pi r^2 h$ where r is the radius and h is the perpendicular height

11

A sphere has radius $2x$ cm

A cone has

radius $3x$ cm

perpendicular height h cm

The sphere and the cone have the same volume.

Work out radius of cone : perpendicular height of cone

Give your answer in the form $a : b$ where a and b are integers.

(4 marks)

PAST PAPERS

Q29**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q5

CH 1 • TOPIC 12**6 marks**

- 5 120 children go on an activity holiday.
The ratio of the number of girls to the number of boys is 3 : 5
- On Sunday, all the children either go sailing or go climbing.
- $\frac{16}{25}$ of the boys go climbing.
- Twice as many girls go sailing as go climbing.
- Work out how many children go sailing on Sunday.

.....

(Total for Question 5 is 6 marks)

Q30**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q5

CH 1 • TOPIC 12 4 marks

- 5 Nancy has some coins with a total value of 85 pence.
She has only 2 pence coins and 5 pence coins.
The ratio

$$\text{number of 2 pence coins} : \text{number of 5 pence coins} = 1 : 3$$

Nancy has more 5 pence coins than 2 pence coins.

How many more?

.....

(Total for Question 5 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE A total

$\div$ sum of parts

ZONE B difference

$\div$ difference in parts

ZONE C one share

$\div$ its parts

- | | |
|----------------------|---|
| 01 Multiplier | new $\div$ old, times everything; greatest number: smallest multiplier. |
| 02 Simplify | Same units; $1 : n$ divide by the first number. |
| 03 Sharing | Value of 1 part (A, B or C), then $\times$ parts. |
| 04 Linking | Make the shared item equal, join the ratios. |
| 05 Problems | Parts $\rightarrow$ value / weight / cost; watch what stays the same. |

BOTTOM LINE Scaling $\rightarrow$ Multiplier • Sharing $\rightarrow$ Value of 1 Part • two ratios $\rightarrow$ make the middle equal.

Answers

Ratio Toolkit · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • RECIPES AND SCALING: THE MULTIPLIER

Q1	Dr Eslam · Multiplier	$450 \times \frac{10}{6} = 750 \text{ g}$
Q2	Practice · Easy	(a) 360 g (b) 25 pancakes
Q3	Practice · Easy	90 g, 450 g, 225 ml, 1.5 onions, 960 ml
Q4	Practice · Easy	23.8 calories
Q5	Practice · Medium	needs 562.5 g butter, 875 g flour, 250 g sugar, 700 g mincemeat, 3 eggs: not enough mincemeat — No
Q6	Practice · Medium	400 ham pizza portions; assumes the sample is representative
Q7	Practice · Hard	£42.28

IDEA 02 • SIMPLIFY AND 1 : n

Q8	Dr Eslam · 1 : n	$40 : 200 = 1 : 5$
Q9	Practice · Easy	1 : 2
Q10	Practice · Easy	$n = 20$
Q11	Practice · Medium	(a) 4 : 3 (b) 480 pixels
Q12	Practice · Medium	36.5 cm
Q13	Practice · Hard	$8 : 2.5 = 3.2 : 1$
Q14	Practice · Very Hard	1 : 2.53

IDEA 03 • SHARING: THE VALUE OF 1 PART

- Q15** Dr Eslam • Zone A 1 part = £40: £80, £120, £160
- Q16** Dr Eslam • Zone B 1 part = 9: $8 \times 9 = 72$
- Q17** Practice • Easy (a) $225 : 475 = 9 : 19$ (b) Daniel £175, Rose £455
- Q18** Practice • Medium (a) raisins 40 g, oats 100 g (b) £1.44
- Q19** Practice • Hard 96
- Q20** Practice • Very Hard 147 euros
- Q21** Edexcel 4MA1 • May 2023 • Paper 1H • Q1 32 goals

IDEA 04 • LINKING TWO RATIOS

- Q22** Dr Eslam • Link 9 : 12 : 10
- Q23** Edexcel 4MA1 • May 2021 • Paper 1H • Q4 35 : 10 : 18: 70 chickens
- Q24** Edexcel 4MA1 • November 2023 • Paper 2H • Q5 18 : 15 : 10: 216 g

IDEA 05 • RATIO PROBLEMS

- Q25** Practice · Medium 240
- Q26** Practice · Hard 31.5 litres
- Q27** Practice · Very Hard $y + x = k(y - x) \Rightarrow y(k - 1) = x(k + 1)$
- Q28** Practice · Very Hard $h = \frac{32}{9}x: 27 : 32$
- Q29** Edexcel 4MA1 · January 2020 · Paper 1H · Q5 57
- Q30** Edexcel 4MA1 · June 2023 · Paper 2H · Q5 10